

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

Ben Geisler

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1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #42. The comparison assesses whether bug fixes or model changes affected results.

Current Comparison:

- **Issue:** #42
- **Baseline:** Commit 607b539 (2025-11-20 12:35:28)
- **Fixed:** Commit cadf81a (2025-11-20 14:17:17)

1.1 Available Snapshot Pairs

Table 1: All Available Snapshot Pairs

	Issue	Title	Baseline Commit	Fixed Commit	Status
1	#42	[P0] Critical Bug: Incorrect death transition calculation in calculate_outcomes.R	607b539	cadf81a	Current
4	#67	[P4] Move snapshot folder	f3674e8	d5d1356	
2	#62	[P4] Move cached file	d046c7c	f3674e8	
3	#64	[P3] impact of bug fixes	8681d1e	8681d1e	

2 Metadata Comparison

Table 2: Metadata Comparison

Item	Baseline	Fixed
Git Commit	607b539	cadf81a
Timestamp	2025-11-20 12:35:28	2025-11-20 14:17:17
R Version	R version 4.5.1 (2025-06-13 ucrt)	R version 4.5.1 (2025-06-13 ucrt)
Issue Number	42	42

2.1 Global Parameters

Table 3: Global Parameter Comparison

	Parameter	Baseline	Fixed	Changed
WTP	WTP	51,000	51,000	No
time_horizon	time_horizon	520	520	No
n_samples	n_samples	5,000	5,000	No
n_sim	n_sim	5,000	5,000	No
cl	cl	0.01923077	0.01923077	No
dr	dr	0.04	0.04	No
DSA_mult	DSA_mult	0.2	0.2	No
USE_BOTH_MODELS	USE_BOTH_MODELS	0	0	No

3 Base Case Results Comparison

Table 4: Base Case Cost-Effectiveness Results Comparison

Strategy	Cost (€)				Effect (QALYs)				ICER (€/QALY)		
	Baseline	Fixed	Difference	% Change	Baseline	Fixed	Difference	% Change	Baseline	Fixed	Difference
control	21,748.34	21,748.34	0.00	0.00	1.8040	1.8040	0.0000	0.00	—	—	—
crp	55,992.46	55,992.46	0.00	0.00	1.9153	1.9153	0.0000	0.00	307,907	307,907	0
tlr	76,439.41	76,439.41	0.00	0.00	1.2957	1.2957	0.0000	0.00	—	—	—
tmb_braf	68,032.43	68,032.43	0.00	0.00	1.8235	1.8235	0.0000	0.00	—	—	—

3.1 Net Monetary Benefit Comparison

Table 5: Net Monetary Benefit Comparison at WTP = €51,000

	Strategy	Baseline (€)	Fixed (€)	Difference (€)
1	control	70,257.69	70,257.69	0.00
2	crp	41,685.57	41,685.57	0.00
4	tmb_braf	24,963.72	24,963.72	0.00
3	tlr	-10,359.72	-10,359.72	0.00

Optimal Strategy: Unchanged (control)

4 PSA Results Comparison

4.1 Summary Statistics

Table 6: PSA Mean Costs and Effects (n = 5,000 simulations)

Strategy	Mean Cost (€)			Mean Effect (QALYs)		
	Baseline	Fixed	Difference	Baseline	Fixed	Difference
control	20,936.34	20,936.34	0.00	1.6999	1.6999	0.0000
crp	54,118.74	54,118.74	0.00	1.6977	1.6977	0.0000
tlr	85,905.64	85,905.64	0.00	1.6230	1.6230	0.0000
tmb_braf	64,950.69	64,950.69	0.00	1.7942	1.7942	0.0000

4.2 95% Credible Intervals

Table 7: PSA 95% Credible Intervals

Strategy	Cost Baseline		Cost Fixed		Effect Baseline		Effect Fixed	
	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%
control	15,783.79	27,129.21	15,783.79	27,129.21	1.1101	2.3569	1.1101	2.3569
crp	40,595.71	69,484.76	40,595.71	69,484.76	1.1276	2.2533	1.1276	2.2533
tlr	61,363.13	115,121.66	61,363.13	115,121.66	1.0532	2.1950	1.0532	2.1950
tmb_braf	47,458.10	85,506.50	47,458.10	85,506.50	1.1067	2.5413	1.1067	2.5413

5 Visual Comparison

5.1 PSA Scatter Plots (Side-by-Side)

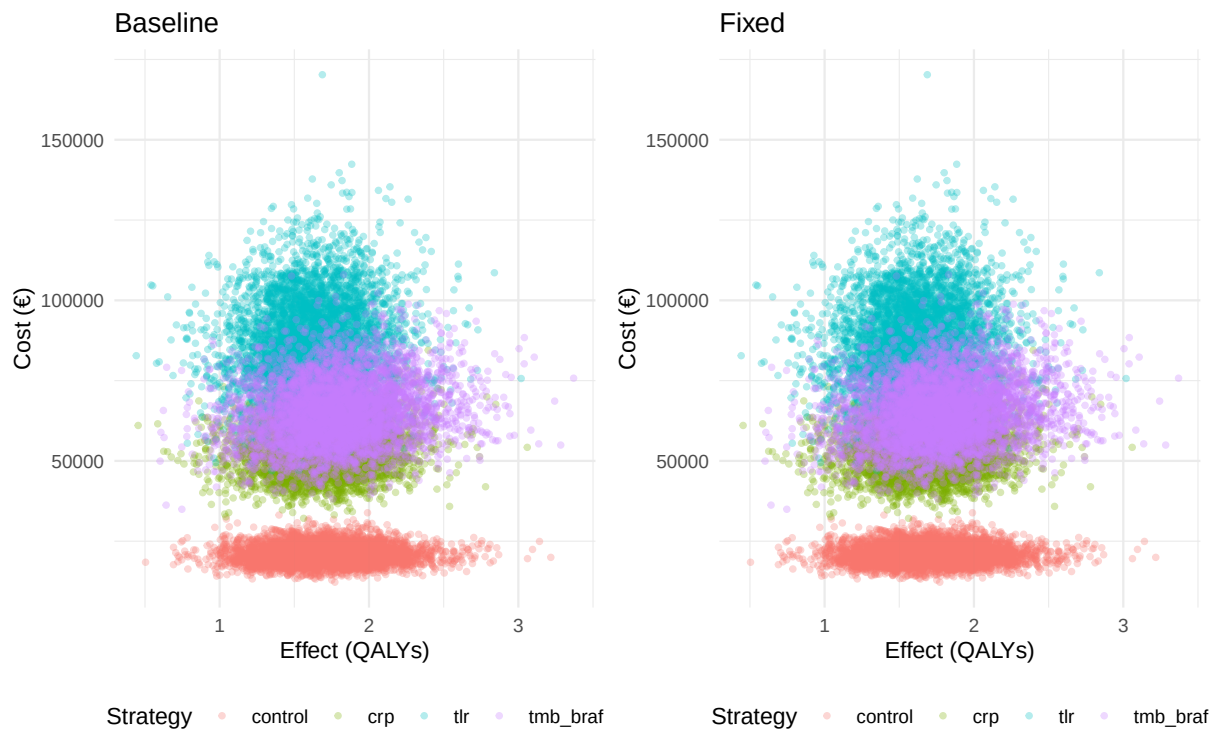


Figure 1: Cost-effectiveness plane showing PSA simulations baseline (left) and fixed (right)

5.2 PSA Scatter Plots (Overlaid)

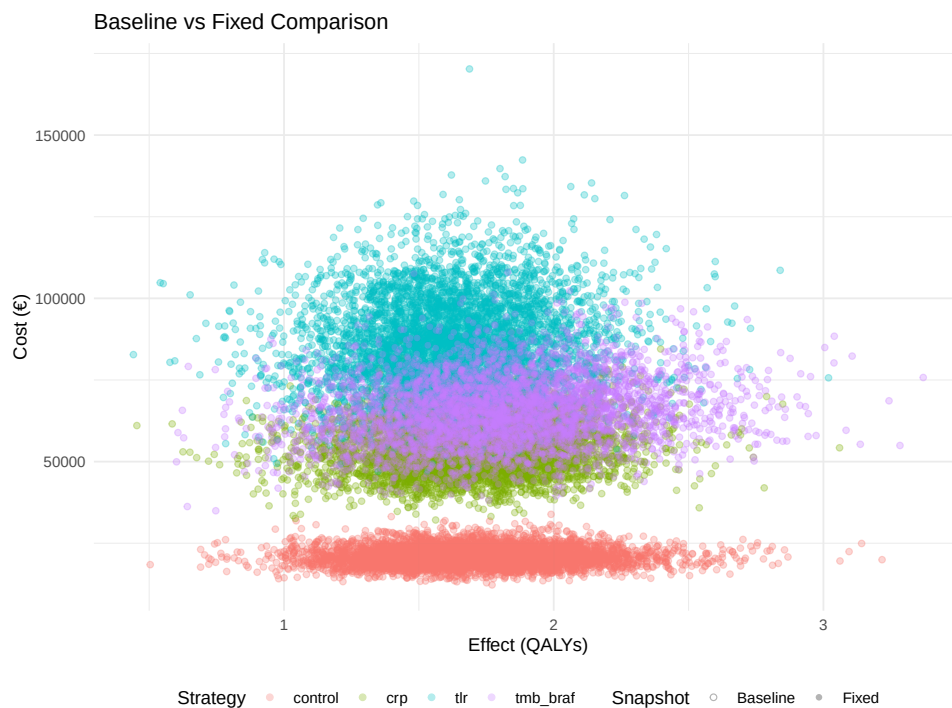


Figure 2: Cost-effectiveness plane with overlaid baseline and fixed PSA simulations

6 Comparison Across Recent Issues

This table summarizes the impact of bug fixes across the 4 most recent issues:

Table 8: Summary of Bug Fix Impacts (4 Most Recent Issues)

Issue	Title	Optimal (Baseline)	Optimal (Fixed)	Changed?	Max Cost Change (€)	Max Effect Change (QALYs)	Max ICER Change (€/QALY)
#42	[P0] Critical Bug: Incorrect death transition calculation in calculate_outcomes.R	control	control	No	0.00	0.0000	0
#67	[P4] Move snapshot folder	control	control	No	0.00	0.0000	0
#62	[P4] Move cached file	control	control	No	0.00	0.0000	0
#64	[P3] impact of bug fixes	control	control	No	0.00	0.0000	0

7 Summary

The comparison for issue #42 shows:

- **Optimal Strategy:** Unchanged (control)
- **Maximum Cost Change:** €0.00 (minor)
- **Maximum Effect Change:** 0.000 QALYs (minor)
- **Maximum ICER Change:** €0/QALY (minor)
- **Parameter Changes:** None detected

Interpretation: Changes are within expected bounds for bug fixes that improve numerical stability or code quality.

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